Tutorial on

Programming Novel AI Accelerators for Scientific Computing

ISC26
June 22, 2026

Murali Emani
Argonne National Laboratory
memani@anl.gov

Varuni Sastry (ANL), Sylvia Howland (Cerebras),
Matthew Povey (SambaNova), Moritz Thüning (Tenstorrent)

Argonne NATIONAL LABORATORY

Tutorial Agenda

https://github.com/argonne-lcf/ALCF-AIAccelerators-tutorial/tree/ISC26

Agenda

Time	Topic	Duration (minutes)	Speaker
2:00 PM - 2:15 AM	Welcome and Overview of the ALCF AI Testbed	15	Murali (ANL)
2:15 PM - 3:15 PM	Cerebras [AI Slides, SDK Slides]	60	Sylvia (Cerebras)
3:15 PM - 4:00 PM	Sambanova [Slides]	45	Matthew (SambaNova)
04:00 PM - 04:30 PM	Coffee Break	30	
04:30 PM - 05:15 PM	Tenstorrent [Slides]	45	Moritz (Tenstorrent)
05:15 PM - 06:00 PM	Hands-on and Q&A	45	Varuni, Murali (ANL)

ALCF AI Testbed

https://www.alcf.anl.gov/alcf-ai-testbed

Cerebras

SambaNova

Groq

Graphcore

Tenstorrent

- Infrastructure of next-generation machines with hardware accelerators customized for artificial intelligence (AI) applications.
- Provide a platform to evaluate usability and performance of machine learning based HPC applications running on these accelerators.
- The goal is to better understand how to integrate AI accelerators with ALCF's existing and upcoming supercomputers to accelerate science insights

ALCF AI Testbed

ALCF AI Testbed Systems are in production and available for allocations to the research community

Training

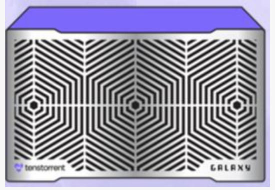
- Cerebras

Cerebras CS-3 – 4 WSE

Inference

- SambaNova SN40L – Metis
- Groq
- Cerebras
- Tenstorrent

4 nodes of 16 SN40L RDUs



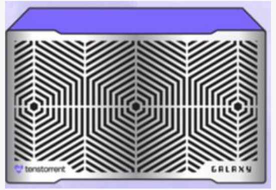
2 Tenstorrent
Galaxy server2 -
32 wormhole chips

Cerebras CS-3 – 4 WSE (upcoming support)

Low-level SDK

- Cerebras
- Tenstorrent

Cerebras CS-3



Tenstorrent Galaxy

Getting Started on ALCF AI Testbed

Available for Allocations

- Cerebras CS-3
- Sambanova Inference – Metis SN40L (Available for all ALCF users via Inference Service Endpoints)
- GroqRack
- Graphcore Bow Pod64

AI Testbed User Guide

Director's Discretionary (DD) awards

- Scaling code
- Preparing for future computing competition
- Scientific computing in support of strategic partnerships.

Allocation Request Form

https://www.alcf.anl.gov/science/directors-discretionary-allocation-program

NAIRR Pilot

Aims to connect U.S. researchers and educators to computational, data, and training resources needed to advance AI research and research that employs AI.

https://nairrpilot.org/

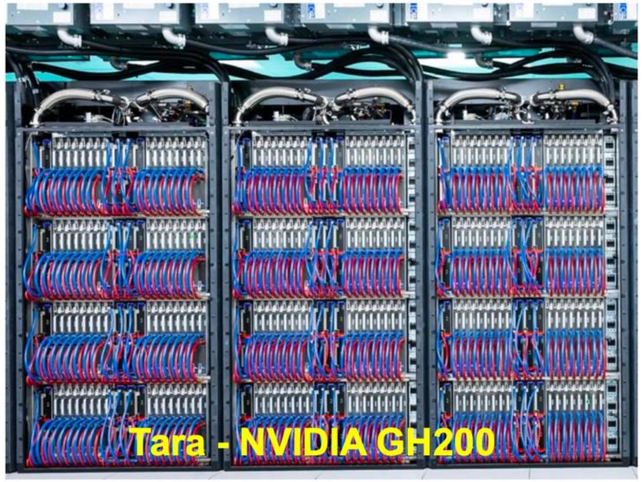
https://docs.alcf.anl.gov/ai-testbed/getting-started/

ALCF AI Testbed

The ALCF AI Testbed houses some of the most advanced AI accelerators for scientific research.
The goal of the testbed is to enable explorations into next-generation machine learning applications and workloads, enabling the ALCF and its user community to help define the role of AI accelerators in scientific computing and how to best integrate such technologies with supercomputing resources.
The AI accelerators complement the ALCF's current and next-generation supercomputers to provide a state-of-the-art computing environment that supports pioneering research at the intersection of AI, big data, and high performance computing (HPC).
The platforms are equipped with architectural features that support AI and data-centric workloads, making them well suited for research tasks involving the growing deluge of scientific data produced by powerful tools, such as supercomputers, light sources, telescopes, particle accelerators, and sensors. In addition, the testbed will allow researchers to explore novel workflows that combine AI methods with simulation and experimental science to accelerate the pace of discovery.

Does this doc need an update? Open an Issue on GitHub

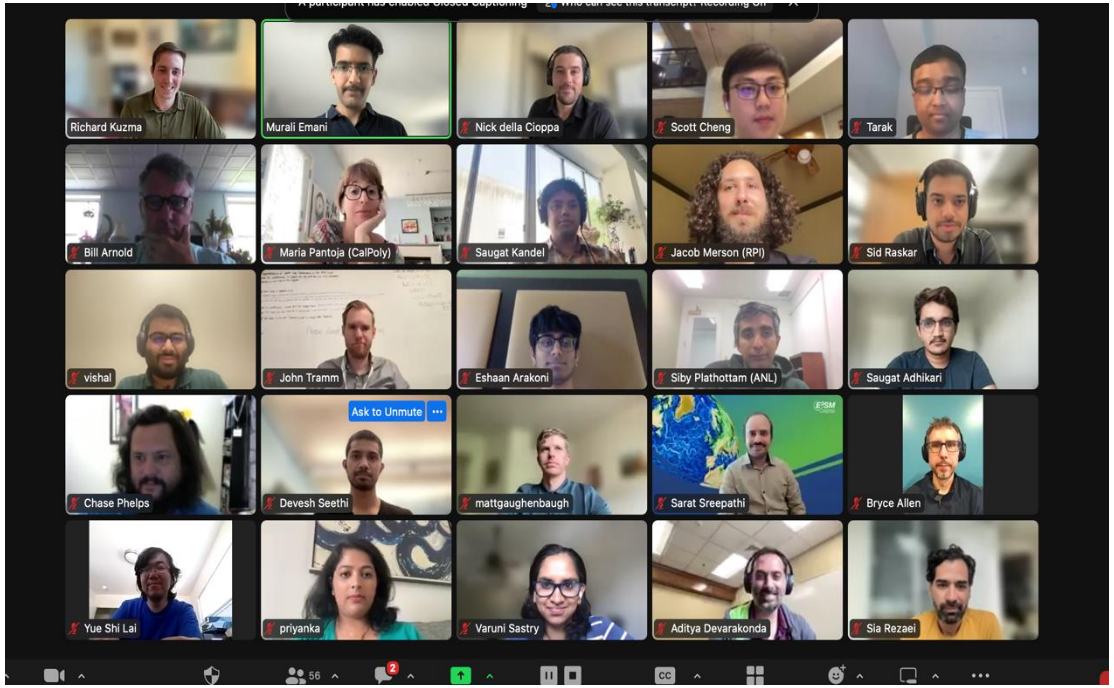
ALCF IS DEPLOYING DIVERSE INFERENCE SYSTEMS FOR SCIENCE



Coming soon:
Tenstorrent Galaxy, Minerva, Tara (Nvidia)

https://docs.alcf.anl.gov/services/inference-endpoints/

AI Testbed Community Engagement



Tutorials at venues such as SC, ISC on "Programming Novel AI accelerators for Scientific Computing"

- NAIRR Annual meeting, SC25, SCA-HPC Asia

Upcoming:

- **ISC26, June 22, 2026**
- Cerebras CS-3 training (TBA)
- SC26 (Nov 26)

- AI training workshops https://www.alcf.anl.gov/ai-testbed-training-workshops

- ATPESC Training

- Introduction to AI-driven Science on Supercomputers

Useful Links

ALCF AI Testbed

- Overview: https://www.alcf.anl.gov/alcf-ai-testbed
- Guide: https://docs.alcf.anl.gov/ai-testbed/getting-started/
- Training:
  - https://www.alcf.anl.gov/ai-testbed-training-workshops
  - https://github.com/argonne-lcf/ALCF-AIAccelerators-tutorial/tree/ISC26
- Allocation Request: https://www.alcf.anl.gov/science/directors-discretionary-allocation-program
- Support: support@alcf.anl.gov

Reach out for any follow up discussions:

Murali Emani (memani@anl.gov)

Varuni Sastry (vsastry@anl.gov), Venkat Vishwanath (Venkat@anl.gov), William Arnold (arnoldw@anl.gov)